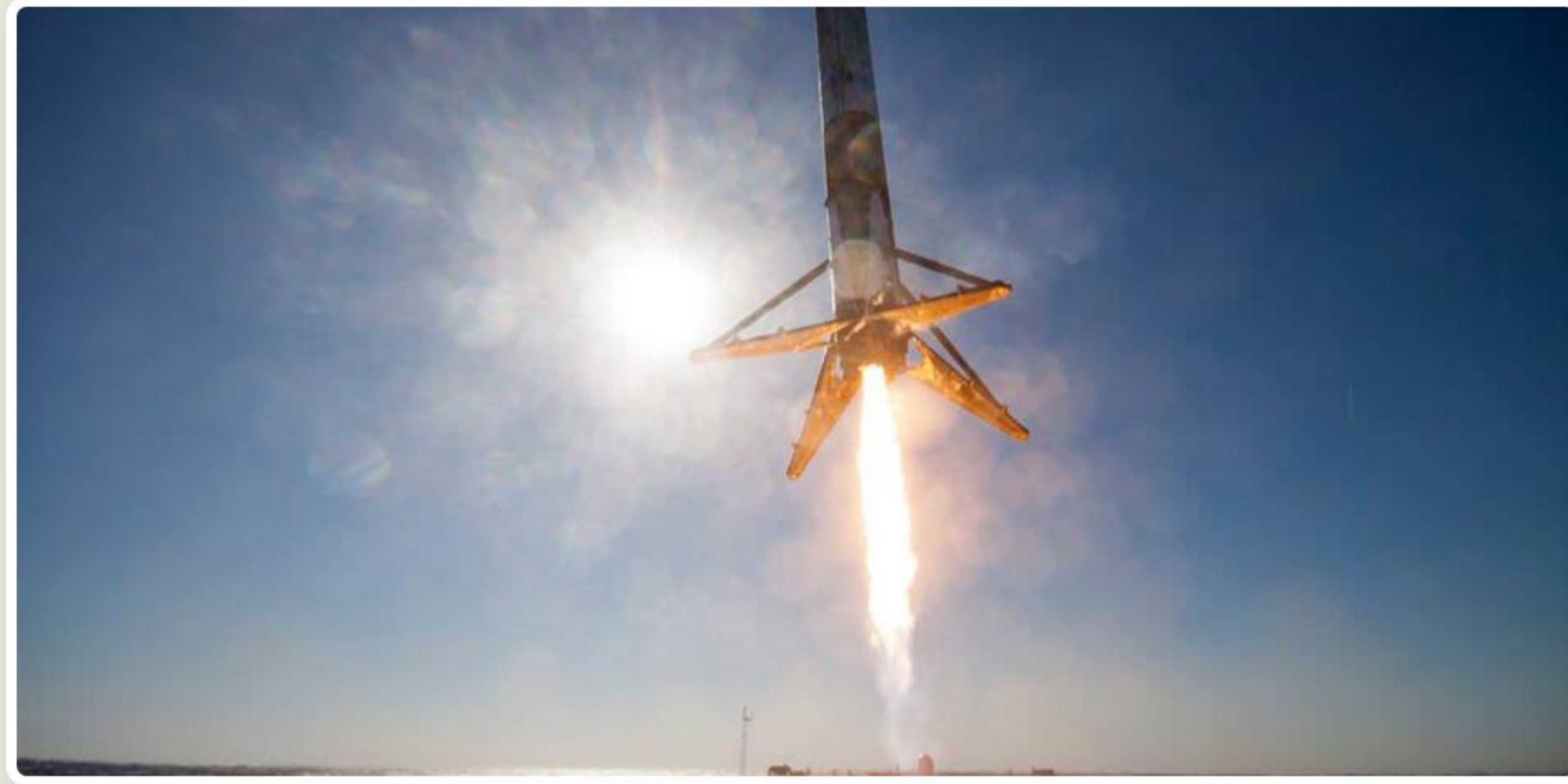




Falcon 9 Landing Prediction

Applying Data Science and Machine Learning to Optimize Launch Cost
Recovery Through First-Stage Predictive Analytics

Executive Summary

This comprehensive study integrates data collection, wrangling, exploratory analysis, and machine learning to predict SpaceX Falcon 9 first-stage landings. By analyzing historical launch data, geographical proximities, and technical specifications, we achieved high-accuracy landing predictions.

- ✓ **Data:** Merged API and Wikipedia records for over 90 launches.
- ✓ **Analysis:** Identified Payload Mass and Orbit Type as critical landing success variables.
- ✓ **Models:** Evaluated Logistic Regression, SVM, Decision Trees, and KNN.
- ✓ **Result:** Decision Tree model achieved the peak training accuracy (0.90).

Methodology: Data Collection



Dual-Source Strategy

We employed a multi-faceted approach to gather a robust dataset, combining official technical metrics with historical contextual data.

- **REST API:** Queried SpaceX API (v4) to extract specific IDs for rockets, payloads, launchpads, and cores.
- **Web Scraping:** Utilized *BeautifulSoup* to parse Wikipedia tables, securing historical launch records not fully detailed in the API.
- **Normalization:** Combined sources into a flat structure for downstream feature engineering.

Data Wrangling & Preparation



Cleaning & Filtering

Initial datasets contained noise and missing values. We focused exclusively on **Falcon 9** launches by removing Falcon 1 and Heavy variants. Null values in the "LandingPad" column were handled, and the target "Class" variable was derived from mission outcomes.



Feature Engineering

Categorical data (Orbits, Launch Sites) were converted using **One-Hot Encoding** to create numeric features for ML models. We standardized the payload mass and flight numbers to ensure uniformity across the predictive algorithms.

Exploratory Data Analysis: SQL

Key statistics queried from the SpaceX SQLite database to identify initial correlations:

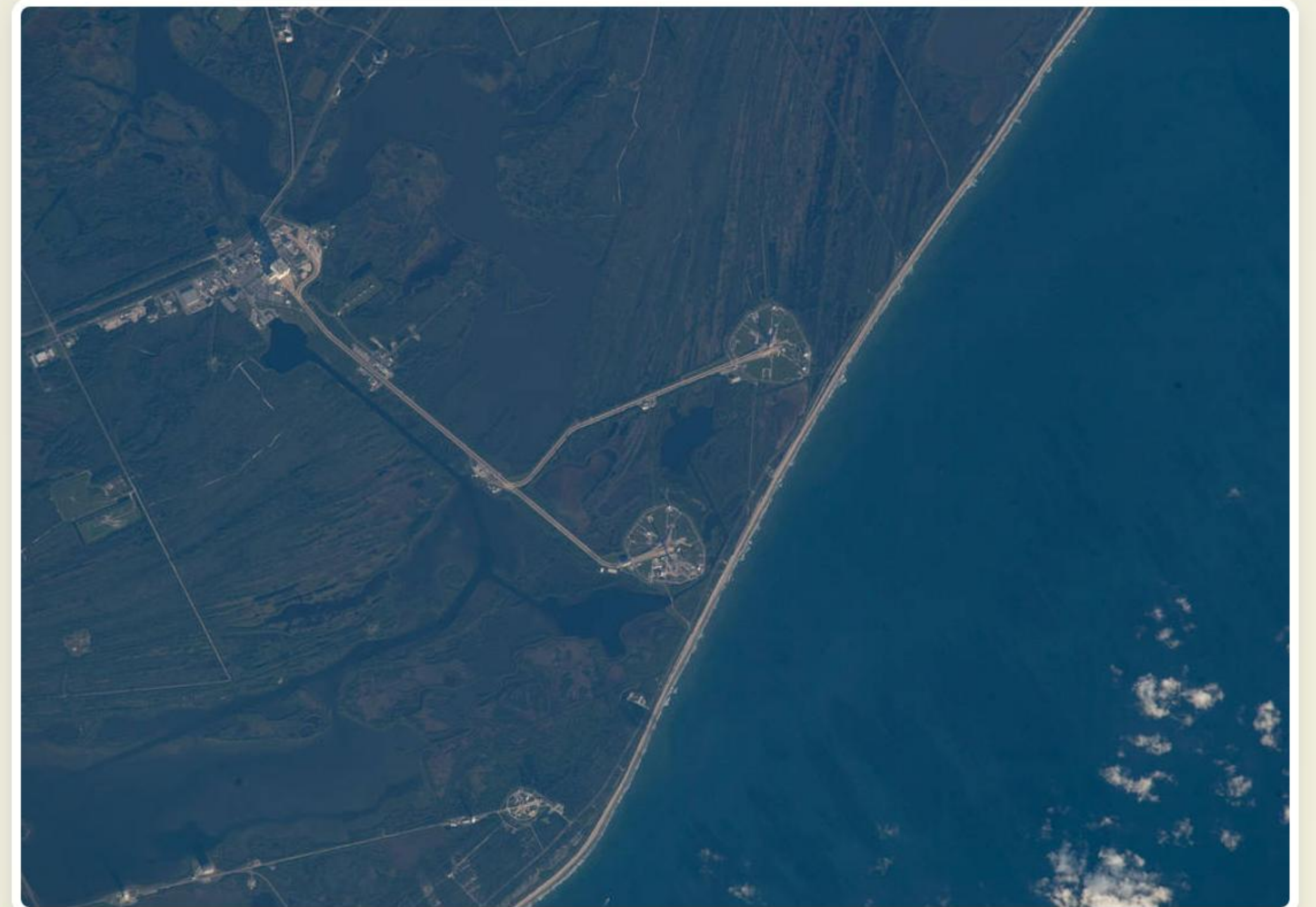
Query Objective	Result / Metric	Insight
Total Payload Mass (NASA CRS)	45,596 kg	High frequency of logistics missions.
Success in Drone Ship (4k-6k kg)	Booster FT B1022, B1026	Successful landings even with heavy loads.
First Successful Ground Pad	2015-12-22	Start of consistent RTLS success.
Most Frequent Success Orbit	GTO (27 launches)	Primary operational orbit for Falcon 9.

Visual Analytics with Folium

Launch Site Proximity

Interactive maps were created to analyze site locations relative to geographical markers:

- **Site Clusters:** Markers for KSC LC-39A, CCAFS SLC-40, and VAFB SLC-4E revealed success density.
- **Coastline Analysis:** All sites are within ~1km of the coast to ensure safe flight trajectories over the ocean.
- **Logistics:** Proximity to highways and railways was verified for equipment transport efficiency.



Interactive Dashboard: Plotly Dash



Real-time Visualization

An interactive tool was developed to allow users to filter and drill down into launch data:

- **Site Dropdown:** Toggle between all sites or specific locations to view success pie charts.
- **Payload Slider:** Adjust the mass range (0 to 10k kg) to see the effect on landing success.
- **Correlation Analysis:** Scatter plots reveal the relationship between payload and outcome for different boosters.

Machine Learning & Prediction

Tuning classification algorithms to determine the probability of a successful first-stage landing based on technical flight parameters.

Classification Model Benchmarking



Note: Scores represent the accuracy on validation data after hyperparameter tuning via GridSearchCV (cv=10).

Key Findings

66%

OVERALL SUCCESS RATE

100%

HEO & SSO SUCCESS

Success Determinants

- **Launch Site:** KSC LC-39A exhibits the highest success probability.
- **Orbits:** ES-L1, GEO, HEO, and SSO have achieved perfect landing rates to date.
- **Payload:** Heavier payloads historically correlated with a higher success rate in recent ASDS landings.

Conclusion & Business Impact

The ability to predict first-stage landing outcomes with **~85% accuracy** provides significant competitive leverage. By anticipating reuse potential, bidding companies can better estimate SpaceX's operational costs—currently **\$62M vs the industry average of \$165M**. Our models confirm that as launch experience increases, the landing success rate improves, cementing reusability as the future of space logistics.

Questions?

SpaceX Falcon 9 Landing Prediction Project

Data Science Capstone Presentation
Applied Machine Learning Methodology

Image Sources



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